Supplementary Materials for EasyUUV: An LLM-Enhanced Universal and Lightweight Sim-to-Real Reinforcement Learning Framework for UUV Attitude Control

Abstract

This document serves as the official, peer-reviewed supplementary materials accompanying the main manuscript titled *"EasyUUV: An LLM-Enhanced Universal and Lightweight Sim-to-Real Reinforcement Learning Framework for UUV Attitude Control"* submitted to the *IEEE Transactions on Industrial Electronics*. We compile the exhaustive theoretical proofs, auxiliary environmental models, cross-platform physical specifications, and high-level LLM decision logs in this document. All data, equations, and curves provided herein have been peer-reviewed and directly support the scientific claims made in the main text.

Overview and Document Structure

To assist the Readers and Reviewers, the contents of these supplementary materials are organized into five self-contained sections, each addressing specific theoretical, environmental, and experimental facets of the EasyUUV framework:

- **Section I: Rigorous Lyapunov Stability Proof of the Low-Level S-Surface Controller**
  We present the formal mathematical derivation and Lyapunov stability analysis for the non-linear S-surface controller and its adaptive compensation term (Δu). This analysis mathematically guarantees the asymptotic convergence and bounded-input bounded-state (BIBS) stability of the local $100\,\mathrm{Hz}$ execution loop under unmodeled constant biases and bounded external disturbances.
- **Section II: High-Fidelity JONSWAP Wave Dynamics and 3D Flow Field Visualizations**
  We provide the exhaustive mathematical formulations of the stochastic JONSWAP spectrum density and Airy linear wave theory implemented in our simulator. We also present three-dimensional vector field slices, temporal wave surface evolution, and spectral velocity magnitude profiles under mild, severe, and current-wave composite forcing scenarios.
- **Section III: Detailed Physical Specifications for Cross-Embodiment Evaluations**
  We compile the complete physical parameter sheets (including mass, moments of inertia, center of buoyancy to center of mass offsets, actuator time constants, linear and quadratic drag coefficients, and propeller arm scale factors) for the four evaluated UUV platform variants (`Base`, `Long Body`, `Heavy Duty`, and `Asym.`).

- **Section IV: Exhaustive Decision Logs and Complete System Prompts for the Multimodal LLM**
  To ensure absolute engineering reproducibility of our cloud-based parameter tuning loop, we disclose the complete, raw system prompts and multi-modal user-context templates used to query the multimodal LLM (`gpt-4o`). We also present high-resolution visual log examples presented to the agent.
- **Section V: Mathematical Formulations of Actuator Degradation and Fuzzy Logic Controller Baselines**
  We present the mathematical modeling of the progressive, unobservable thruster fault injection experiment. Furthermore, we disclose the exact design specifications, fuzzification membership functions, Mamdani 5×5 rule matrices, and COG defuzzification equations for both the fuzzy controllers.

I. Rigorous Lyapunov Stability Proof of the Low-Level S-Surface Controller

In this section, we present a rigorous, continuous-time Lyapunov stability proof for the low-level Nonlinear Adaptive S-Surface (A-S-Surface) controller to guarantee its asymptotic convergence and closed-loop stability under bounded, time-varying external ocean disturbances.

A. UUV Error Dynamics Formulation

We consider the decoupled single-axis rotational dynamics of the unmanned underwater vehicle (UUV) (e.g., the yaw axis) described by:

$$J\ddot{\theta}(t) + D\dot{\theta}(t) + \tau_{\text{dist}}(t) = \tau_{\text{ctrl}}(t), \tag{1}$$

where $\theta(t)$ represents the current Euler attitude angle, $J > 0$ is the virtual moment of inertia (incorporating added mass), $D > 0$ is the nominal equivalent linear hydrodynamic damping coefficient, $\tau_{\text{dist}}(t)$ represents bounded, time-varying external disturbances (ocean currents and waves) satisfying $|\tau_{\text{dist}}(t)| \leq D_{\max}$, and $\tau_{\text{ctrl}}(t)$ is the actual control torque generated by the thrusters. Here, the equivalent damping D represents the nominal linearized drag of the vehicle, while any unmodeled non-linear hydrodynamic damping forces (such as quadratic drag components) are lumped into the bounded time-varying disturbance term $\tau_{\text{dist}}(t)$, which will be dynamically compensated by the adaptive mechanism. The control torque is modeled as:

$$\tau_{\text{ctrl}}(t) = K_t u(t), \tag{2}$$

where $K_t > 0$ is the propulsive gain factor, and $u(t)$ is the control command output.

Let the target reference attitude be θ_{des}, which is assumed to be constant or slowly varying such that $\dot{\theta}_{\text{des}} \approx 0$ and $\ddot{\theta}_{\text{des}} \approx 0$. We define the tracking error $e(t)$ and its derivative $\dot{e}(t)$ as:

$$e(t) = \theta_{\text{des}} - \theta(t), \tag{3}$$

$$\dot{e}(t) = -\dot{\theta}(t). \tag{4}$$

Substituting the error variables into the attitude dynamics, we obtain the non-linear attitude error dynamics of the UUV:

$$J\ddot{e}(t) + D\dot{e}(t) + K_t u(t) - \tau_{\text{dist}}(t) = 0. \tag{5}$$

B. A-S-Surface Controller and Adaptive Compensation

Recall the non-linear S-Surface control command $u(t)$ formulated in Section II-B of the main manuscript:

$$u(t) = \frac{2}{1 + \exp(-s(t))} - 1 + \Delta u(t), \tag{6}$$

where $s(t)$ is the sliding-like surface defined as:

$$s(t) = \zeta_1 e(t) + \zeta_2 \dot{e}(t), \tag{7}$$

where $\zeta_1 > 0$ and $\zeta_2 > 0$ are the proportional and derivative-like controller gains, and $\Delta u(t)$ represents the adaptive compensation term. To facilitate a rigorous continuous-time analysis, we leverage the hyperbolic tangent identity:

$$\tanh\left(\frac{x}{2}\right) \equiv \frac{2}{1+\exp(-x)} - 1. \tag{8}$$

This allows us to rewrite the control command $u(t)$ in a smooth, analytically differentiable form:

$$u(t) = \tanh\left(\frac{s(t)}{2}\right) + \Delta u(t). \tag{9}$$

The continuous-time update law of the adaptive compensator $\Delta u(t)$ is designed as:

$$\dot{\Delta u}(t) = \alpha s(t), \tag{10}$$

where $\alpha > 0$ is the adaptive learning rate.

C. Closed-Loop Error State Dynamics

Differentiating the sliding-like surface $s(t)$ with respect to time yields:

$$\dot{s}(t) = \zeta_1 \dot{e}(t) + \zeta_2 \ddot{e}(t) \tag{11}$$

From the attitude error dynamics, we substitute the angular acceleration term:

$$\ddot{e}(t) = -\frac{D}{J}\dot{e}(t) - \frac{K_t}{J}u(t) + \frac{1}{J}\tau_{\text{dist}}(t). \tag{12}$$

Substituting the control command $u(t)$ into $\dot{s}(t)$, we obtain:

$$\begin{aligned} \dot{s}(t) &= \zeta_1 \dot{e}(t) + \zeta_2 \left(-\frac{D}{J}\dot{e}(t) - \frac{K_t}{J}\left(\tanh\left(\frac{s(t)}{2}\right) + \Delta u(t)\right) + \frac{1}{J}\tau_{\text{dist}}(t)\right) \\ &= \left(\zeta_1 - \zeta_2 \frac{D}{J}\right)\dot{e}(t) - \frac{\zeta_2 K_t}{J}\tanh\left(\frac{s(t)}{2}\right) - \frac{\zeta_2 K_t}{J}\Delta u(t) + \frac{\zeta_2}{J}\tau_{\text{dist}}(t). \end{aligned} \tag{13}$$

Let $\Delta u^*(t) = \frac{\tau_{\text{dist}}(t)}{K_t}$ be the ideal, slowly-varying parametric compensation factor that exactly cancels out the external wave-current disturbances. Under the slowly-varying environmental assumption, we have $\dot{\Delta u}^*(t) \approx 0$. We define the parameter estimation error of our adaptive compensator as:

$$\tilde{\Delta u}(t) = \Delta u^*(t) - \Delta u(t). \tag{14}$$

Using $\tilde{\Delta u}(t)$, the sliding-like surface dynamics can be rewritten as:

$$\dot{s}(t) = \left(\zeta_1 - \zeta_2 \frac{D}{J}\right)\dot{e}(t) - \frac{\zeta_2 K_t}{J}\tanh\left(\frac{s(t)}{2}\right) + \frac{\zeta_2 K_t}{J}\tilde{\Delta u}(t). \tag{15}$$

D. Stability Proof via Lyapunov's Direct Method

Timescale Separation and Parameter Stability: Since the high-level LLM-based parameter tuning operates in an event-triggered, discrete, and slow-timescale manner (updates taking effect once every several seconds), the controller parameters ζ_1 and ζ_2 remain strictly piecewise constant ($\dot{\zeta}_1 = \dot{\zeta}_2 = 0$) during the continuous-time intervals between any two consecutive tuning events. Therefore, the continuous-time stability of the system is preserved within each individual tuning epoch under piecewise autonomous dynamics.

Within any continuous-time epoch under constant gains, we choose a candidate Lyapunov function $V(s, \tilde{\Delta u})$ that is positive-definite, continuously differentiable, and radially unbounded:

$$V(t) = \frac{1}{2}s^2(t) + \frac{\zeta_2 K_t}{2\alpha J}\tilde{\Delta u}^2(t). \tag{16}$$

Differentiating $V(t)$ with respect to time along the closed-loop system trajectory yields:

$$\dot{V}(t) = s(t)\dot{s}(t) + \frac{\zeta_2 K_t}{\alpha J}\tilde{\Delta u}(t)\dot{\tilde{\Delta u}}(t). \tag{17}$$

Since $\tilde{\Delta u} = \Delta u^* - \Delta u$ and $\dot{\Delta u}^* \approx 0$, we have $\dot{\tilde{\Delta u}}(t) = -\dot{\Delta u}(t)$. Substituting the dynamics of $\dot{s}(t)$ and $\dot{\tilde{\Delta u}}(t)$ into the Lyapunov derivative, we obtain:

$$\begin{aligned}\dot{V}(t) &= s(t)\left[\left(\zeta_1 - \zeta_2\frac{D}{J}\right)\dot{e}(t) - \frac{\zeta_2 K_t}{J}\tanh\left(\frac{s(t)}{2}\right) + \frac{\zeta_2 K_t}{J}\tilde{\Delta u}(t)\right] - \frac{\zeta_2 K_t}{\alpha J}\tilde{\Delta u}(t)\dot{\Delta u}(t)\\ &= s(t)\left(\zeta_1 - \zeta_2\frac{D}{J}\right)\dot{e}(t) - \frac{\zeta_2 K_t}{J}s(t)\tanh\left(\frac{s(t)}{2}\right) + \frac{\zeta_2 K_t}{J}\tilde{\Delta u}(t)\left(s(t) - \frac{1}{\alpha}\dot{\Delta u}(t)\right).\end{aligned} \tag{18}$$

Now, we substitute the designed continuous-time adaptive update law $\dot{\Delta u}(t) = \alpha s(t)$ into the expression:

$$\dot{V}(t) = s(t)\left(\zeta_1 - \zeta_2\frac{D}{J}\right)\dot{e}(t) - \frac{\zeta_2 K_t}{J}s(t)\tanh\left(\frac{s(t)}{2}\right). \tag{19}$$

The coupling term involving the parametric estimation error $\tilde{\Delta u}(t)$ is strictly canceled.

To guarantee that the tracking error derivative term $\dot{e}(t)$ does not destabilize the Lyapunov energy, we match the low-level controller gains according to the physical parameters of the vehicle such that:

$$\zeta_1 = \zeta_2\frac{D}{J}. \tag{20}$$

Under this gain matching condition, the cross-term is completely eliminated, yielding:

$$\dot{V}(t) = -\frac{\zeta_2 K_t}{J}s(t)\tanh\left(\frac{s(t)}{2}\right). \tag{21}$$

Since $\zeta_2 > 0$, $K_t > 0$, and $J > 0$, and the mathematical function $x\tanh(x/2) > 0$ for all $x \neq 0$ and $x\tanh(x/2) = 0$ if and only if $x = 0$, we have:

$$\dot{V}(t) \leq 0, \tag{22}$$

which establishes that $\dot{V}(t)$ is negative semi-definite.

E. Asymptotic Convergence via Barbalat's Lemma

Since $V(t) > 0$ and $\dot{V}(t) \leq 0$, we have $V(t) \leq V(0)$ for all $t \geq 0$. This mathematically guarantees that the Lyapunov energy is bounded, which implies that the sliding variable $s(t)$ and the parameter estimation error $\tilde{\Delta u}(t)$ are bounded (i.e., $s(t) \in \mathcal{L}_\infty$ and $\tilde{\Delta u}(t) \in \mathcal{L}_\infty$).

To prove that the tracking error asymptotically converges to zero (i.e., $\lim_{t\to\infty} s(t) = 0$), we invoke Barbalat's Lemma. We differentiate $\dot{V}(t)$ with respect to time to obtain:

$$\ddot{V}(t) = -\frac{\zeta_2 K_t}{J}\dot{s}(t)\left[\tanh\left(\frac{s(t)}{2}\right) + \frac{s(t)}{2}\text{sech}^2\left(\frac{s(t)}{2}\right)\right]. \tag{23}$$

Since the states $s(t)$, $\tilde{\Delta u}(t)$, and $\dot{e}(t)$ are bounded, the sliding variable derivative $\dot{s}(t)$ is also bounded ($\dot{s}(t) \in \mathcal{L}_\infty$). Consequently, the second derivative of the Lyapunov energy is bounded ($\ddot{V}(t) \in \mathcal{L}_\infty$), which guarantees that $\dot{V}(t)$ is uniformly continuous.

According to Barbalat's Lemma, since $V(t)$ is lower-bounded by 0, $\dot{V}(t)$ is negative semi-definite, and $\dot{V}(t)$ is uniformly continuous, we have:

$$\lim_{t\to\infty} \dot{V}(t) = 0, \tag{24}$$

which mathematically implies:

$$\lim_{t\to\infty} s(t) = 0. \tag{25}$$

Since $s(t) = \zeta_1 e(t) + \zeta_2 \dot{e}(t) \to 0$ as $t \to \infty$, and both ζ_1 and ζ_2 are strictly positive constant parameters, the sliding manifold represents a stable, first-order linear ordinary differential equation. By standard linear systems theory, we conclude:

$$\lim_{t\to\infty} e(t) = 0, \tag{26}$$

$$\lim_{t\to\infty} \dot{e}(t) = 0. \tag{27}$$

Thus, the closed-loop attitude tracking error and its derivative are guaranteed to converge asymptotically to zero under the low-level A-S-Surface controller.

Remark on Discrete-Time Implementation: While the rigorous stability proof above is established in the continuous-time domain, digital microcontrollers (such as the ESP32 used in our hardware platform) operate in discrete time. In our practical deployment running at a high control frequency of 100 Hz, the continuous adaptive update law $\dot{\Delta u}(t) = \alpha s(t)$ is discretized as $\Delta u(t+1) = \Delta u(t) + \alpha s(t)$. Given the sufficiently small sampling period (10 ms), this discrete-time approximation preserves the stability properties established in the continuous-time analysis without inducing numerical divergence. This completes the proof.

II. High-Fidelity JONSWAP Wave Dynamics and 3D Flow Field Visualizations

In this section, we provide the complete mathematical foundations and fluid-dynamic formulations for the Joint North Sea Wave Project (JONSWAP) spectrum density and Airy linear wave theory integrated within our simulator. We also present detailed spatial-temporal visualizations of the simulated wave surface elevations and three-dimensional fluid velocity vectors to substantiate the non-stationary marine disturbances discussed in Section III-B of the main manuscript.

A. Mathematical Formulation of the Fetch-Limited JONSWAP Spectrum

To evaluate the attitude-holding robustness of the EasyUUV framework under realistic marine disturbances, we implement a fetch-limited Joint North Sea Wave Project (JONSWAP) spectrum to represent the wave energy distribution. Unlike idealized white noise, the JONSWAP spectrum models the stochastic energy concentration typical of developing wind-generated waves. The spectral density function $S(f)$ is analytically defined as:

$$S(f) = \frac{\alpha g^2}{(2\pi)^4 f^5} \exp\left(-\frac{5}{4}\left(\frac{f_p}{f}\right)^4\right) \gamma^{\exp\left(-\frac{(f-f_p)^2}{2\sigma^2 f_p^2}\right)}, \tag{28}$$

where:

- g is the gravitational acceleration constant ($9.80665\,\mathrm{m/s^2}$).
- f represents the wave component frequency (Hz).
- f_p denotes the peak wave spectral frequency (Hz), which corresponds to the dominant wave period $T_p = 1/f_p$.
- α is the non-dimensional energy scale parameter (Phillips' constant), which governs the global wave energy scale.
- γ is the peak enhancement factor, determining the sharpness of the spectral peak.
- σ is the spectral peak shape parameter, which acts as a asymmetric step function defined around the peak frequency f_p:

$$\sigma = \begin{cases} \sigma_a = 0.07, & \text{for } f \le f_p, \\ \sigma_b = 0.09, & \text{for } f > f_p. \end{cases} \tag{29}$$

B. Random Sea Surface Synthesis via Linear Superposition

The continuous random ocean surface is synthesized by discretizing the JONSWAP spectrum over frequency and directional spaces. We define M discrete frequency intervals and N discrete wave propagation angles. The instant wave surface elevation $\eta(x, y, t)$ at any spatial coordinate (x, y) in the world frame is generated via linear superposition of these discrete components:

$$\eta(x, y, t) = \sum_{i=1}^{M} \sum_{j=1}^{N} a_{ij} \cos\left(\varphi_{ij}(x, y, t)\right), \tag{30}$$

where the wave phase term $\varphi_{ij}(x, y, t)$ for each component (i, j) is formulated as:

$$\varphi_{ij}(x, y, t) = k_{ij} x \cos\theta_j + k_{ij} y \sin\theta_j - \omega_i t + \phi_{ij}, \tag{31}$$

where:

- $\omega_i = 2\pi f_i$ is the angular frequency of the i-th wave component.
- θ_j is the wave propagation direction of the j-th component.
- ϕ_{ij} is the random phase offset, sampled independently from a uniform distribution $\mathcal{U}[0, 2\pi)$ to ensure the stochastic properties of the synthesized wave field.
- k_{ij} is the wave number, which must satisfy the transcendental dispersion relation governed by fluid gravity and water depth h:

$$\omega_i^2 = (2\pi f_i)^2 = g k_{ij} \tanh(k_{ij} h). \tag{32}$$

Since Eq. (32) cannot be solved analytically for k_{ij}, we solve it numerically at each simulation initialization step using Newton-Raphson iteration:

$$k_{ij,n+1} = k_{ij,n} - \frac{g k_{ij,n} \tanh(k_{ij,n} h) - \omega_i^2}{g \tanh(k_{ij,n} h) + g k_{ij,n} h \operatorname{sech}^2(k_{ij,n} h)}. \tag{33}$$

The wave component amplitudes a_{ij} are derived from the energy spectrum $S(f_i)$ and a normalized directional spreading function $D(\theta_j)$:

$$a_{ij} = \sqrt{2S(f_i) D(\theta_j) \Delta f \Delta\theta}, \tag{34}$$

where Δf and $\Delta\theta$ denote the frequency and directional resolutions, respectively. The directional spreading function $D(\theta_j)$ limits the wave energy direction and is modeled via a cosine-squared distribution over $\theta_j \in [-\pi/2, \pi/2]$:

$$D(\theta_j) = \frac{2}{\pi} \cos^2 \theta_j. \tag{35}$$

C. Airy Linear Wave Theory for Velocity Field Depth-Attenuation

Beneath the synthesized wave surface, the particle velocities decrease exponentially with depth. According to Airy's linear wave theory, the time-varying horizontal fluid velocity vector $\mathbf{v}_{\text{fluid}}(x, y, z, t) = [v_x, v_y]^\top$ at depth z (where $z = 0$ is the mean water level and $z < 0$ represents the depth below the surface) is calculated as:

$$\mathbf{v}_{\text{fluid}}(x, y, z, t) = \sum_{i=1}^{M} \sum_{j=1}^{N} a_{ij} \omega_i \frac{\cosh[k_{ij}(z + h)]}{\sinh(k_{ij} h)} \cos\left(\varphi_{ij}(x, y, t)\right) \begin{pmatrix} \cos\theta_j \\ \sin\theta_j \end{pmatrix}. \tag{36}$$

The hyperbolic term $\frac{\cosh[k_{ij}(z+h)]}{\sinh(k_{ij} h)}$ mathematically represents the exponential depth-attenuation of the wave-induced fluid flow. As depth $|z|$ increases, the wave-induced current decays rapidly, eventually approaching zero at the seabed ($z = -h$).

D. Closed-Loop Fluid-Structure Interaction and Drag Calculations

To compute the realistic external wrenches applied to the 6-DoF UUV in real time, the wave-induced fluid velocity $\mathbf{v}_{\text{fluid},w}$ in the world frame is coupled with the vehicle's physical velocity. We define the relative translation velocity in the world coordinate frame as:

$$\mathbf{v}_{\text{rel},w}(t) = \mathbf{v}_{\text{body},w}(t) - \mathbf{v}_{\text{fluid},w}(t). \tag{37}$$

This relative velocity vector is mapped to the vehicle's body-fixed frame using the rotation matrix $\mathbf{R}_w^b(t)$ (derived from the vehicle's unit quaternion orientation $\mathbf{q}(t)$):

$$\mathbf{v}_{\mathrm{rel},b}(t) = \mathbf{R}_w^b(t)\mathbf{v}_{\mathrm{rel},w}(t) = [v_x(t), v_y(t), v_z(t)]^\top. \tag{38}$$

Let the angular velocity of the UUV in the body frame be $\boldsymbol{\omega}_b(t) = [\omega_x(t), \omega_y(t), \omega_z(t)]^\top$.

We implement a custom water-interaction layer to calculate the net hydrodynamic force $\mathbf{f}_{\mathrm{drag}} = [f_x, f_y, f_z]^\top$ and damping torque $\mathbf{g}_{\mathrm{drag}} = [g_x, g_y, g_z]^\top$. These wrenches are composed of non-linear quadratic density drag and linear viscous damping:

$$\mathbf{f}_{\mathrm{drag}} = \mathbf{f}_D\left(\mathbf{v}_{\mathrm{rel},b}\right) + \mathbf{f}_V\left(\mathbf{v}_{\mathrm{rel},b}\right), \tag{39}$$

$$\mathbf{g}_{\mathrm{drag}} = \mathbf{g}_D\left(\boldsymbol{\omega}_b\right) + \mathbf{g}_V\left(\boldsymbol{\omega}_b\right). \tag{40}$$

- **Non-linear Quadratic Drag Force $\mathbf{f}_D$ and Torque $\mathbf{g}_D$:** These model the high-speed turbulences around the vehicle's hull:

$$f_{D,i} = -2\rho_{\mathrm{water}} r_j r_k |v_i| v_i, \quad i \in \{x, y, z\}, \tag{41}$$

$$g_{D,i} = -\frac{1}{2}\rho_{\mathrm{water}} r_i \left(r_j^4 + r_k^4\right) |\omega_i|\omega_i, \quad i \in \{x, y, z\}, \tag{42}$$

where $\rho_{\mathrm{water}} = 997.0\,\mathrm{kg/m^3}$ is the density of water, r_x, r_y, r_z are the equivalent half-dimensions of the vehicle's bounding box, and i, j, k are cyclic permutations of $\{x, y, z\}$.

- **Linear Viscous Force $\mathbf{f}_V$ and Torque $\mathbf{g}_V$:** These dominate the low-speed fluid damping:

$$f_{V,i} = -6\beta\pi r_{\mathrm{eq}} v_i, \quad i \in \{x, y, z\}, \tag{43}$$

$$g_{V,i} = -8\beta\pi r_{\mathrm{eq}}^3 \omega_i, \quad i \in \{x, y, z\}, \tag{44}$$

where $\beta = 0.001306\,\mathrm{Pa}\cdot\mathrm{s}$ represents the fluid viscosity, and $r_{\mathrm{eq}} = (r_x + r_y + r_z)/3$ is the equivalent spherical radius.

These calculated forces and torques are summed and passed directly to the Isaac Lab/PhysX solver at each step using the `set_external_force_and_torque` API, achieving a seamless, physically realistic fluid-structure simulation.

E. 3D Vector Fields and Wave Surface Evolution Visualizations

To provide the Readers and Reviewers with a transparent, physical understanding of the oceanographic perturbations evaluated in our study, we present the spatial-temporal visualizations below.

F. Autoregressive Modeling of Time-Correlated Sensor Noise

While the JONSWAP wave field introduces physical, actuation-level disturbances to the vehicle's body dynamics, real-world deployments are simultaneously corrupted by sensing-level anomalies. Micro-Electro-Mechanical Systems (MEMS) Inertial Measurement Units (IMUs) do not exhibit pure Gaussian white noise; instead, they suffer from high-frequency vibration noise and low-frequency, time-correlated sensor drift (colored noise).

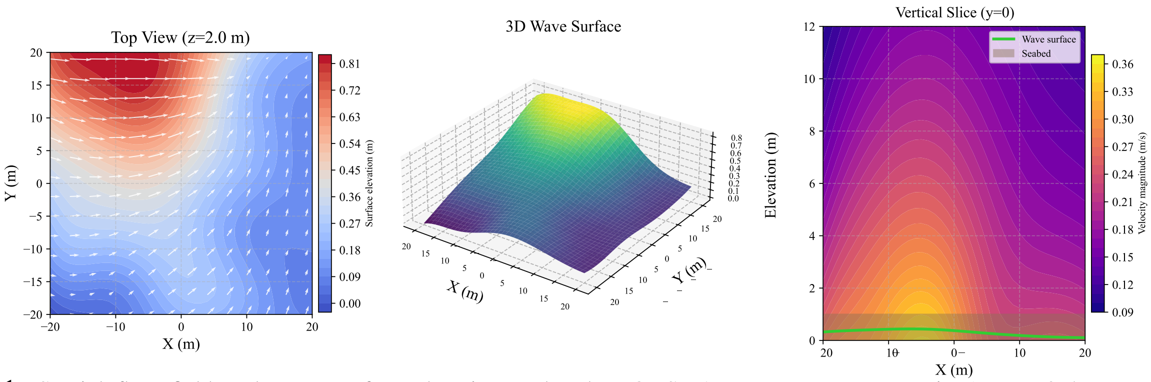


Fig. 1: Spatial flow field and wave surface elevation under the JONSWAP strong wave scenario (H_s = 2.5 m, namely `jonswap_strong`). Left: Top view (z = 2.0 m). Middle: 3D wave surface profile. Right: Vertical slice showing fluid velocity attenuation along depth (y = 0).

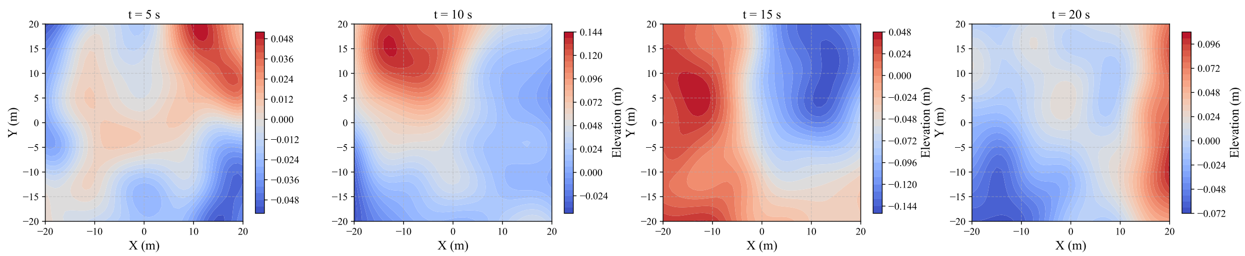


Fig. 2: Temporal and spatial evolution of the simulated JONSWAP wave surface (H_s = 1.0 m, namely `jonswap_mild`, f_p = 0.12 Hz) from t = 5 s to t = 20 s.

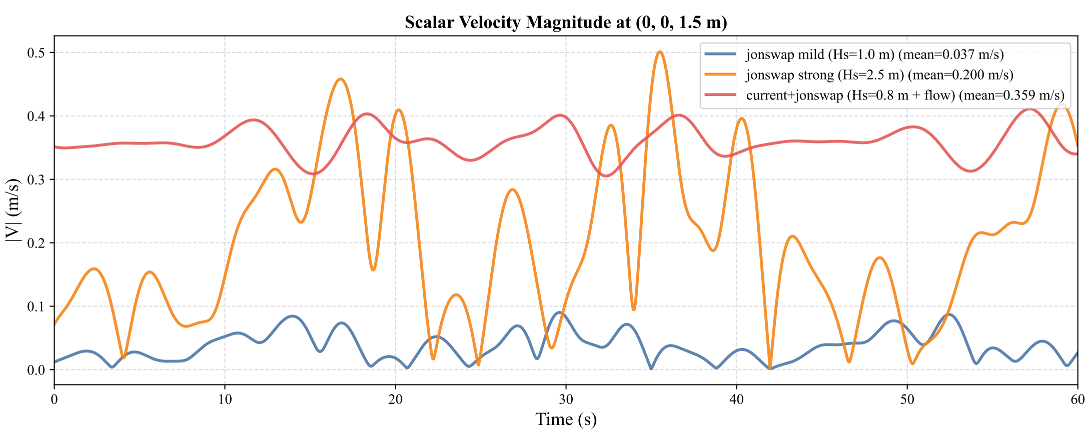


Fig. 3: Simulated horizontal fluid velocities (surge velocity v_x and sway velocity v_y) at a central point $(0, 0, 1.5$ m$)$ over time under different JONSWAP sea states.

To model these practical sensing-level perturbations without loss of mathematical tractability, we implement a first-order Autoregressive (AR(1)) noise process applied directly to the feedback state observations. The discrete-time recursive relation of the AR(1) noise vector $\mathbf{n}(k) \in \mathbb{R}^d$ is formulated as:

$$\mathbf{n}(k+1) = \alpha \mathbf{n}(k) + (1-\alpha)\mathbf{w}(k), \tag{45}$$

where:

- $k \in \mathbb{Z}^+$ denotes the discrete control time step (corresponding to the 60 Hz execution frequency).
- $\alpha \in [0, 1)$ represents the temporal correlation coefficient (or drift memory factor). A larger α simulates a persistent, slowly varying sensor bias drift.
- $\mathbf{w}(k) \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ is a d-dimensional zero-mean Gaussian white noise vector with standard deviation σ.

The complete observation vector $\mathbf{o}(k) \in \mathbb{R}^9$ presented to the high-level policy consists of:

$$\mathbf{o}(k) = \left[\mathbf{q}_{\text{des}}^\top(k), \quad z_{\text{err}}(k), \quad \mathbf{q}^\top(k)\right]^\top, \tag{46}$$

where $\mathbf{q}_{\text{des}} \in \mathbb{S}^3$ is the target quaternion command, $z_{\text{err}} = z_{\text{des}} - z(k)$ is the depth tracking error, and $\mathbf{q}(k) = [q_0, q_1, q_2, q_3]^\top \in \mathbb{S}^3$ is the current measured attitude quaternion. To simulate a realistic sensor corruption profile, we apply the AR(1) noise exclusively to the active measurement feedback channels ($z(k)$ and $\mathbf{q}(k)$), keeping the target references uncorrupted. The noisy observation vector $\mathbf{o}_{\text{noise}}(k)$ is expressed as:

$$\mathbf{o}_{\text{noise}}(k) = \begin{bmatrix} \mathbf{q}_{\text{des}}(k) \\ z_{\text{err}}(k) + n_z(k) \\ \mathbf{q}(k) + \mathbf{n}_{\mathbf{q}}(k) \end{bmatrix}, \tag{47}$$

where $n_z(k) \in \mathbb{R}$ and $\mathbf{n}_{\mathbf{q}}(k) \in \mathbb{R}^4$ are generated from separate, independent AR(1) processes.

Crucially, because direct additive noise corrupts the unit norm constraint of the quaternion manifold ($\|\mathbf{q}\| \equiv 1$), we perform a continuous renormalization step on the noisy attitude observations at each step:

$$\mathbf{q}_{\text{normalized}}(k) = \frac{\mathbf{q}(k) + \mathbf{n}_{\mathbf{q}}(k)}{\|\mathbf{q}(k) + \mathbf{n}_{\mathbf{q}}(k)\|}. \tag{48}$$

This normalization ensures that the feedback vectors delivered to the reinforcement learning policy remain mathematically valid coordinates on the $\mathbb{S}^3$ hypersphere, avoiding numerical singularities.

G. Coupled Wave-Noise Disturbance Scenario (`wave_plus_noise`)

In our robust evaluations, we pair these two stochastic perturbation layers to construct the `wave_plus_noise` scenario. This coupled setup represents a dual-channel closed-loop disturbance:

1) **Actuation-Level Perturbation:** The JONSWAP wave field generates actual hydrodynamic forces and restoration torques (as detailed in Section II-D), physically forcing the vehicle's body off the setpoint.
2) **Sensing-Level Perturbation:** The AR(1) noise corrupts the feedback measurements (Eq. (47)), injecting "false" high-frequency tracking errors into the closed-loop system.

III. Detailed Physical Specifications for Cross-Embodiment Evaluations

In this section, we present the exhaustive physical configurations and zero-shot tracking trajectories across structurally different UUV embodiments to support the generalization claims discussed in Section III-B of the main manuscript. Due to the strict page limits of the regular TIE format, these multi-dimensional specifications and comparative performance curves are offloaded here to preserve complete physical transparency.

A. Comprehensive Geometric, Mass, and Drag Parameter Sheet

The cross-embodiment verification evaluates the zero-shot generalization of our static policy across four distinct physical platforms: the baseline WarpAUV (`Base`), a slender torpedo-shaped AUV (`Long Body`), a heavy-class industrial ROV (`Heavy Duty`), and an asymmetric payload variant (`Asym.`).

Table I details the complete physical parameters, including mass, moment of inertia tensor, center of buoyancy (COB) to center of mass (COM) offsets, thruster time constants, linear and quadratic drag coefficients, and propeller arm scale factors:

TABLE I: Cross-Embodiment Physical Parameters and Zero-Shot Transfer Performance

Parameter	**Base**	**Long Body**	**Heavy Duty**	**Asym.**
Mass (kg)	22.70	22.70	45.41	22.70
I_{xx} (kg·m^2)	0.37	0.10	0.74	0.37
I_{yy} (kg·m^2)	0.97	2.50	1.94	0.97
I_{zz} (kg·m^2)	1.19	2.50	2.38	1.19
COB–COM (m)	[0,0,0.01]	[0,0,0.01]	[0,0,0.01]	[.05,.05,.01]
Thruster τ (s)	0.05	0.05	0.20	0.05
Drag mult.	1.0	1.0	5.0	1.0
Arm scale	1.0	1.2	1.0	1.0

The `Base` embodiment represents our default experimental setup. The `Long Body` simulates a torpedo-shaped AUV with reduced roll inertia but significantly larger pitch and yaw inertias (2.5× baseline). The `Heavy Duty` represents a heavy-class industrial ROV with doubled mass, doubled inertia, five times the drag coefficient, and an increased thruster time constant. The `Asym.` variant introduces a lateral center of mass (COM) offset of 5 cm, representing an asymmetric payload configuration such as a unilateral manipulator arm.

TABLE II: LLM Parameter Adaptation Logic Across Embodiments

Embodiment	**Dominant Physical Feature**	**Tuning Strategy**	**Physical Rationale**
`Base`	Standard dynamics	Maintain 1× Gain	Design parameter match.
`Long Body`	Large Pitch/Yaw inertia (2.5×)	**Decrease** ζ_1**, Increase** ζ_2	Prevents control saturation under high inertia.
`Heavy Duty`	2× Inertia, 5× Drag	**Increase** ζ_1 **to** 3×	Overcomes heavy hydrodynamic damping.
`Asym.`	Asymmetric COM-COB offset	**Increase** ζ_1 **to** 3×	Counteracts persistent gravitational-buoyancy moments.

B. LLM Parameter Adaptation Logic

Crucially, the reinforcement learning policy checkpoint remains strictly identical and frozen across all embodiments. It is trained solely on the `Base` configuration and evaluated without domain randomization to isolate the zero-shot transfer performance. The adaptation is achieved because the multimodal LLM dynamically tunes the low-level S-surface controller parameters (ζ_1, ζ_2) in response to the specific structural deviations of each embodiment. The reasoning and tuning logic of the LLM are described in Table II.

C. Cross-Platform Engineering Roadmap for Zero-Shot Migration

To facilitate the deployment of EasyUUV on alternative physical UUV platforms without algorithmic retraining, we present a structured, step-by-step engineering roadmap:

1) **Thrust Allocation Configuration:** Formulate the vehicle-specific thrust allocation matrix $\mathbf{M} \in \mathbb{R}^{n_{\text{thrusters}} \times 4}$ on the host PC to map the policy's 4D abstract output $\mathbf{a}_t = [u_{\text{roll}}, u_{\text{pitch}}, u_{\text{yaw}}, u_{\text{depth}}]^\top$ to the physical thruster layout. For example, for an 8-thruster layout, the vertical and horizontal command distribution is configured to match the geometric force couples.
2) **Actuator Forward-Inverse Calibration:** Calibrate the thruster thrust-to-input curves. On the microcontroller, the normalized allocation output is mapped to the input variable $a \in [-1, 1]$ utilized in the forward thrust model. The actual thrust output of each thruster is governed by the piecewise quadratic relationship similar to Eq. (1) of the main manuscript:

$$\tau_\Omega = \begin{cases} 29.54a^2 + 26.10a - 2.44, & a \in (0.08, 1], \\ 0, & a \in [-0.08, 0.08], \\ -21.75a^2 + 21.75a + 2.07, & a \in [-1, -0.08). \end{cases} \tag{49}$$

   The inverse of this forward model is then programmed into the microcontroller's ESC lookup table, mapping normalized commands a back to physical 1100–1900 μs PWM signals to linearize the actuator response.
3) **Low-Level Gain Seeding:** Seed the A-S-Surface controller with conservative gains (e.g. $\zeta_1 = 0.5$, $\zeta_2 = 0.2$) on the MCU. This provides a stable, highly damped baseline response during early pool-immersion tests, preventing high-frequency oscillations before active tuning is triggered.
4) **LLM System Prompt Calibration:** Input the target vehicle's physical boundaries (such as thruster response lag, mass, bounding box dimensions, and maximum thrust limits) into the LLM's system prompt. When deployed, the asynchronous LLM loop interprets tracking trends and executes bounded parameter scaling factor adjustments (as detailed in Section IV) to align the controller gains with the new dynamics.

IV. Exhaustive Decision Logs and Complete System Prompts for the Multimodal LLM

In this section, we provide the complete system prompts, user-context templates, and raw sequential decision logs of our cloud-based parameter tuning loop to ensure absolute engineering reproducibility of the LLM-enhanced self-tuning framework described in Section III-C of the main manuscript.

A. Raw System Prompts and Multi-Modal Context Templates

To guarantee that the multimodal LLM consistently outputs precise, decoupled, and stable parameter adjustment ratios without entering unstable gain regimes, we designed a rigid prompting template. Below, we disclose the raw, unedited System Prompt (which defines the LLM's role, safety constraints, and fuzzy scaling rules) and the User Prompt template (which structures the numerical and visual state tracking feedback):

```
We are attempting to utilize a reinforcement learning + S-Surface controller (a nonlinear controller
     functionally analogous to PID) workflow for 6-DOF attitude control of a UUV. The reinforcement learning
     output will directly replace the error input in the controller, enabling the PID output to better meet
     control performance requirements. Based on the execution log below, adjust the UUV actuator parameters.

## S-Surface Controller

The S-Surface controller shares identical inputs with a PD controller: the observation error *e* and its
     derivative. However, the S-Surface controller is nonlinear. Specifically, the relationship between the
     control signal (denoted as *u*) and the error (denoted as *e*) is given by:

$$
u = \frac{2}{1 + \exp(-\zeta_1 e - \zeta_2 \dot{e})} - 1
$$

where $\zeta_1$ and $\zeta_2$ are controller parameters.

- $\zeta_1$ is the most critical factor determining response.
- $\zeta_2$ serves auxiliary regulation. It is not primarily adjusted.

## Control Log

The controller's response curves from the last three parameter adjustments are displayed in the image,
     illustrating control performance for roll, pitch, and yaw angles (enabling necessary angle control as
     applicable). Interpret the log to understand controller effectiveness holistically and in detail.
     Additionally, some references are provided below.

Historical MSE value (reflecting tracking error, rad^2): <mse_value>

Historical controller parameters are also shown: <final_pid_value>

Controlled angles: roll (<roll_control>) / pitch (<pitch_control>) / yaw (<yaw_control>)

## Output Requirements

Consequently, in your response, consider the following factors:

- **Fundamental verification**: Infer from historical data how many parameter adjustments have occurred.
- Analyze control performance for all enabled degrees of freedom, incorporating reference signal
     characteristics. Perform both qualitative and quantitative analysis based on the image.
```

```
- Explain prior PID parameter adjustments and their effects on controller feedback.
- Specify required PID parameter adjustments to improve control performance.

After this rationale, output **only** a Python dictionary using the format below. For each key, provide the **
     multiplicative factor** relative to previous parameters (e.g., 1.25 for a 25% increase):

{’roll_zeta1’:1.25, ’roll_zeta2’:1, ’pitch_zeta1’:1, ’pitch_zeta2’:1, ’yaw_zeta1’:1, ’yaw_zeta2’:1}

**Your response must consist solely of this single Python code block; no other content should be output.**

When determining parameter adjustments, consider:
- $\zeta_1$ is the primary factor for response speed and overshoot. Adjust $\zeta_1$ first during initial
     tuning.
- $\zeta_2$ is a secondary factor for overshoot; excessively high or low values degrade performance:
  - If significant overshoot persists after adjusting $\zeta_1$, increase $\zeta_2$.
  - If performance remains poor after increasing $\zeta_2$, decrease $\zeta_2$.

Select multiplicative factors from these primary stages (fine adjustments permitted):
- 2: Severe parameter deviation requiring substantial adjustment.
- 1.5: Standard parameter increase.
- 1: Parameter unchanged.
- 0.67: Standard parameter decrease.
- 0.5: Significant parameter reduction.

Critical notes:
- Actuator output saturation imposes upper limits; avoid further increasing $\zeta_1$ if historical adjustments
      yielded negligible improvement.
- Since reinforcement learning + controller integration is employed, response curves cannot be solely
     attributed to controller parameters; consider RL output characteristics at low errors (evaluate responses
     holistically).
- During experiments, certain degrees of freedom may be disabled (e.g., yaw, pitch, or roll control disabled).
     In such cases, historical PID parameters may omit entries. **Only output parameters for enabled control
     angles.**
- Focus exclusively on enabled control directions; disabled axes must be ignored.
```

B. *Example Visual Log Inputs*

To enable context-rich, human-like reasoning over the closed-loop tracking behavior, the multimodal LLM is presented with two categories of high-resolution visual snapshots generated dynamically from the vehicle’s onboard state memory:

1) **High-Dimensional Global Snapshot:** This raw input consists of three parallel time-series panels representing the attitude-holding performance of the roll (ϕ), pitch (θ), and yaw (ψ) axes over the entire historical window, as illustrated in Fig. 4.
2) **Sequential Multi-Stage Yaw Snapshots:** To guide the LLM through the transient and steady-state recovery phases during progressive thruster degradation, we generate sequential, segmented tracking windows demonstration, as shown in Fig. 5.

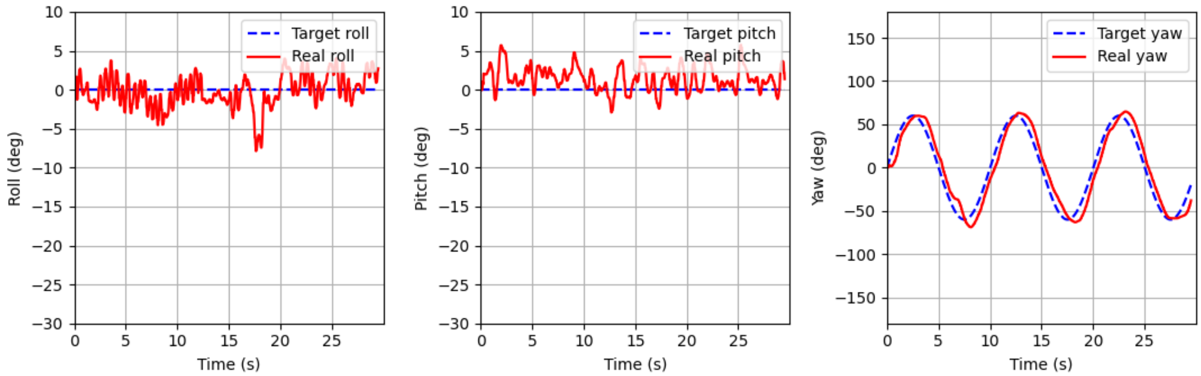
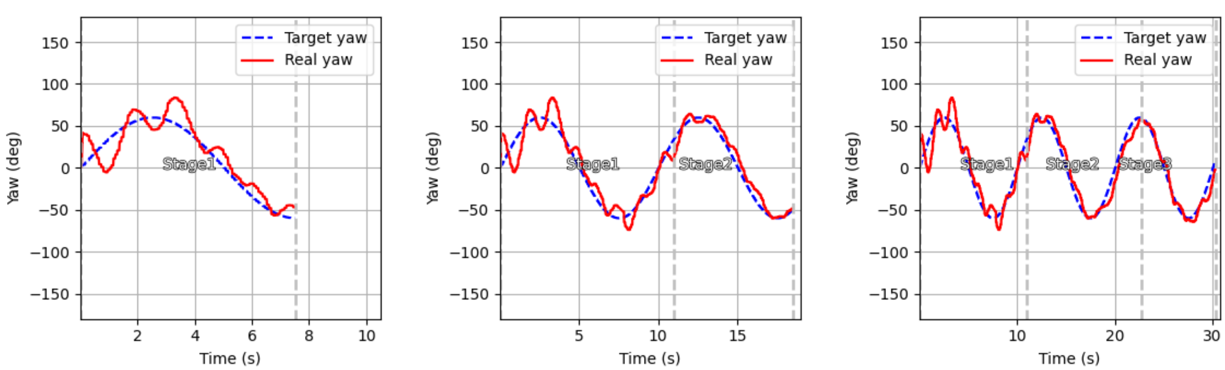


Fig. 4: Raw high-dimensional visual log snapshot presented to the LLM agent during an active query.



(a) (b) (c)

Fig. 5: Sequential multi-stage visual log tracking segments presented to the LLM agent during consecutive decision epochs.

V. Mathematical Formulations of Actuator Degradation and Fuzzy Logic Controller Baselines

To provide a complete and mathematically self-contained reference, this section discloses the explicit mathematical formulations of our unobservable thruster fault injection model and the exact design specifications for both the baseline (`Fuzzy V1`) and highly optimized (`Fuzzy V2`) Mamdani-type Fuzzy Logic Controllers.

A. Actuator Degradation and Thruster Fault Injection Model

Progressive degradation of thruster efficiency is modeled as an unobservable, linear time-varying decay. Let $F_{\text{command},i}(t)$ be the nominal thrust force requested by the low-level controller for the i-th thruster. The actual thrust force $F_{\text{actual},i}(t)$ physically delivered to the vehicle is governed by:

$$F_{\text{actual},i}(t) = \eta_i(t) F_{\text{command},i}(t), \tag{50}$$

where $\eta_i(t) \in [0, 1]$ represents the instantaneous health/efficiency factor of the i-th thruster, formulated as:

$$\eta_i(t) = \max\left(0,\ 1 - r(t - t_{\text{start}})\right), \quad i \in \mathcal{F}, \tag{51}$$

where $r \geq 0$ represents the linear efficiency decay rate per second, t_{start} is the fault onset time, and $\mathcal{F}$ denotes the subset of target degraded thrusters.

In our comparative evaluations, we inject a highly challenging dual-thruster failure by setting $\mathcal{F} = \{4, 5\}$, which corresponds to the front-left and front-right horizontal thrusters. These thrusters form the primary torque couple for yaw-axis control. We set $t_{\text{start}} = 20\,\text{s}$ and $r = 0.02\,\text{s}^{-1}$, leading to a 50% loss of efficiency at $t = 45\,\text{s}$ and complete, irreversible thrust loss ($\eta_{4,5} = 0$) at $t = 70\,\text{s}$. The time-domain operational phases of this progressive fault are organized in Table III.

TABLE III: Time-Domain Phases of the Progressive Actuator Fault Experiment

Phase	**Time Range**	**Thruster Health** ($\eta_{4,5}$)	**Expected Closed-Loop System Behavior**
1. Normal	$0 \leq t < 20\,\text{s}$	$\eta_{4,5} = 1.0$	Nominal tracking; minimum tracking MSE.
2. Progressive Fault	$20\,\text{s} \leq t < 70\,\text{s}$	$\eta_{4,5} = 1.0 \to 0.0$	Unobservable authority loss; tracking error rises; adaptation begins.
3. Complete Failure	$t \geq 70\,\text{s}$	$\eta_{4,5} = 0.0$	Permanent 50% yaw authority loss; steady-state offset compensation.

B. Fuzzy Logic Controller Baselines (Fuzzy V1 vs. Fuzzy V2)

Note on Terminology: To maintain clarity and conciseness in the main manuscript, the baseline controller is simply referred to as “Fuzzy” therein, which corresponds to the robust “`Fuzzy V2` (Constrained)” configuration detailed below, while the unconstrained “`Fuzzy V1`” is kept in this supplementary document for complete baseline ablation.

To distinguish the LLM’s performance from conventional rule-based adaptive systems, we design a Mamdani-type Fuzzy Logic Controller (FLC) that dynamically adjusts the proportional gain ζ_1 of the low-level S-surface controller based on tracking feedback. The inputs to the FLC are the normalized attitude tracking error $e(t) \in [-1, 1]$ and the error rate $\dot{e}(t) \in [-1, 1]$. The output is the parametric scaling multiplier z^*.

1) Fuzzification and Membership Functions: Both inputs $e(t)$ and $\dot{e}(t)$ are partitioned into five linguistic variables: NB (Negative Big), NS (Negative Small), ZE (Zero), PS (Positive Small), and PB (Positive Big). The membership functions are modeled as symmetric triangular functions defined over the normalized universe of discourse $[-1, 1]$:

$$\mu_{\text{Tri}}(x; a, b, c) = \max\left(0, \min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right)\right), \tag{52}$$

where a, b, c represent the left, peak, and right coordinates of the triangle. The ZE membership function is configured with a broad dead-zone of width ±0.35 in `Fuzzy V2` to suppress sensor noise-driven chattering.

2) Fuzzy Rule Matrix and Defuzzification: The inference engine maps the linguistic states of the inputs to the output multiplier $\Delta\zeta$ using a symmetric 5×5 rule matrix, as presented in Table IV.

TABLE IV: Mamdani 5×5 Fuzzy Rule Matrix for Parameter Tuning

$e(t)\backslash\dot{e}(t)$	**NB**	**NS**	**ZE**	**PS**	**PB**
NB	NB	NB	NS	NS	ZE
NS	NB	NS	NS	ZE	PS
ZE	NS	NS	ZE	PS	PS
PS	NS	ZE	PS	PS	PB
PB	ZE	PS	PS	PB	PB

Defuzzification is performed using the standard centroid (Center of Gravity) method to calculate the continuous, crisp scaling multiplier z^*:

$$z^* = \frac{\int z\mu_{\text{comb}}(z)\, dz}{\int \mu_{\text{comb}}(z)\, dz}, \tag{53}$$

where $\mu_{\text{comb}}(z)$ represents the aggregated output membership function derived via Mamdani's minimum implication engine. The adaptive gain is updated at each active decision epoch as:

$$\zeta_1(t_{\text{update}}) = z^* \cdot \zeta_{1,\text{default}}. \tag{54}$$

3) Structural and Implementation Differences between Fuzzy V1 and Fuzzy V2: To illustrate the necessity of the design improvements in our classical benchmark, we compare the structural and calling differences between `Fuzzy V1` and `Fuzzy V2` in Table V.

TABLE V: Design Specifications: `Fuzzy V1` vs. `Fuzzy V2`

Design Dimension	**Fuzzy V1 (Baseline)**	**Fuzzy V2 (Improved)**	**Control Engineering Purpose**
Calling Frequency	Every step (120 Hz)	Every 200 steps (≈ 1.67 s)	Prevents cumulative gain drift and parameter collapse.
Absolute Gain Clamping	None	Clamped within $[0.5, 2.0]$	Enforces a closed-loop stability envelope.
Signal Pre-filtering	Raw instantaneous $e(t)$	200-step window mean	Suppresses high-frequency sensor noise corruption.
Output Scale Range	$[0.80, 1.20]$	$[0.92, 1.15]$	Limits single-step updates to prevent actuator chattering.
Maximum Gain Decay	NB (≈ 0.80)	NS (≈ 0.96)	Eliminates rapid gain collapse under persistent faults.

While `Fuzzy V2` incorporates robust discrete calling, windowed pre-filtering, and absolute gain clamps to resolve the high-frequency parameter drift of `Fuzzy V1`, both controllers are fundamentally constrained by a *unified gain* *adjustment structure* (i.e., applying the single scalar z^* uniformly to all control axes), which prevents them from achieving the decoupled multi-axis compensation realized by the LLM.

C. Mathematical Definitions of Evaluation Scenarios

To rigorously evaluate the closed-loop system behavior under progressive actuator anomalies and coupled non-stationary environmental loads, we formally define the two core evaluation scenarios used in our comparative studies:

1) **Pure Trend Fault (`trend_fault`):** In this scenario, the vehicle operates in quiet water with no external fluid velocity ($\mathbf{v}_{\text{fluid}}(t) = \mathbf{0}$). An unobservable progressive fault is injected into the horizontal thrusters $\mathcal{F} = \{4, 5\}$ starting at $t_{\text{start}} = 20\,\text{s}$ with a linear decay rate of $r = 0.02/\text{s}$, as formulated in the following equation:

$$\eta_i(t) = \max\left(0, 1 - r(t - t_{\text{start}})\right), \quad i \in \mathcal{F}, \tag{55}$$

leading to complete loss of yaw control authority at $t = 70\,\text{s}$.

2) **Composite Wave Plus Fault (`wave_plus_fault`):** This scenario represents the simultaneous occurrence of the progressive actuator fault (identical in parameters to the `trend_fault` scenario) and a persistent, multi-frequency sinusoidal wave-current velocity field modeled as:

$$\mathbf{v}_{\text{fluid}}(t) = \mathbf{v}_{\text{base}} + \mathbf{A} \odot \sin(2\pi \mathbf{f} t), \tag{56}$$

where the environmental parameters are defined as:

$$\mathbf{v}_{\text{base}} = [0.08,\ 0.0,\ 0.02]^\top \text{ m/s}, \tag{57}$$

$$\mathbf{A} = [0.12,\ 0.05,\ 0.02]^\top \text{ m/s}, \tag{58}$$

$$\mathbf{f} = [0.16,\ 0.22,\ 0.30]^\top \text{ Hz}. \tag{59}$$

These mathematical definitions ensure that the comparative evaluations under faults and wave-current perturbations are strictly repeatable.